YOUTUBE CHANNEL: S KHAN ACADEMY

MTH643 CURRENT PAPER FINAL FILE

SOLVED BY SANAULLAH KHAN

CONTACT NO:03066793776

Subject Enrolment

1. Online classes are available (Mathematics)
2. Solved graded activities
   - Assignment's
   - Quizzes
   - GDB's
3. Solved quiz files
4. Past papers
5. Any info about virtual university of Pakistan

Final project (MTH620)

Email:

Sanaullahkhan03066793@gmail.com

Facebook:

Sanaullah Khan

Instagram:

Mr.khan.1431

YouTube:

S KHAN ACADEMY

SUBSCRIBE MY CHANNEL

CLICK HERE:

https://youtube.com/channel/UCcicgZ0tjL8Nc_P_KB2YNVg

SANAULLAH KHAN
MSc. MATHEMATICS
FOR QUIZ CONTECT
(03066793776)

YOUTUBE CHANNEL: S KHAN ACADEMY

1 **Question**

Find sum and write program $\sum_{k=1}^{n}\left(\frac{3}{x}\right)^{k}$ and take the value as an input from user then run the program by giving n as 5

CODE:

```
clc
clear all
close all
syms x
n=input('Enter the value of n: ');
sum=0;
for k=1:n

 sum=sum+(3/x)^k;

end
disp(sum)
```

2 **Question**

Find the numerical solution of the following equation using ode45 command and plot the curve on [-2 2].

$$y'(x) = y - 3x^2 + 7 \qquad y(0) = 1$$

CODE:

```
clc
clear all
close all
f=@(x,y)y-3*x^2+7;
interval=[-2,2];
y0=1;
[x,y]=ode45(f,interval,y0)
plot(x,y)
```

SANAULLAH KHAN
MSc. MATHEMATICS
FOR QUIZ CONTECT
(03066793776)

YOUTUBE CHANNEL: S KHAN ACADEMY

3 **Question**

$\int_1^2 (cos(x) - logx + e^x)dx$ **solve the equation using Simpson's 1/8 and keep 10 sub intervals.**

CODE:

```
clc
clear all
close all
f=@(x)(cos(x)-log(x)+exp(x));
a=1;
b=2;
n=10;
h=(b-a)/n;
sumofodd=0;
sumofeven=0;
for k=1:n-1
 x(k)=a+k*h;
 if mod(k,2)==0
 sumofeven=sumofeven+f(x(k));
 else
 sumofodd=sumofodd+f(x(k));
 end
end
answer=h/3*(f(a)+f(b)+2*sumofeven+4*sumofodd)
```

4 **Question**

Find the numerical solution of the following equation using ode45

command and plot the curve on [0 2].

$$y'(x) = y - x^2 + 1 \qquad y(0) = 0.5$$

SANAULLAH KHAN
MSc. MATHEMATICS
FOR QUIZ CONTECT
(03066793776)

YOUTUBE CHANNEL: S KHAN ACADEMY

CODE:

```
clc
clear all
close all
f=@(x,y)y-x^2+1;
interval=[0,2];
y0=0.5;
[x,y]=ode45(f,interval,y0)
plot(x,y)
```

5 **Question**

Find the numerical solution of the following equation using ode45 command and plot the curve on [0 2].

$$y'(x) = y - x^2 + 1 \qquad y(0) = 0.5{:}\,0.3{:}\,3$$

CODE:

```
clc
clear all
close all
f=@(x,y)y-x^2+1;
interval=[0,2];
y0=0.5:0.3:3;
[x,y]=ode45(f,interval,y0)
plot(x,y)
```

6 **Question**

Find the numerical solution of the following equation using ode45 command and plot the curve on x=0:0.2:2

$$y'(x) = y - x^2 + 1 \qquad y(0) = 0.5$$

SANAULLAH KHAN
MSc. MATHEMATICS
FOR QUIZ CONTECT
(03066793776)

YOUTUBE CHANNEL: S KHAN ACADEMY

CODE:

```
clc
clear all
close all
f=@(x,y)y-x^2+1;
interval=0:0.2:2;
y0=0.5;
[x,y]=ode45(f,interval,y0)
plot(x,y)
```

7 **Question**

Write a for loop to add first 20 numbers which are divisible by 5.

CODE:

```
clc
clear all
close all
sum=0;
for i=5:5:100;
sum=sum+i;
 end
disp(sum)
```

8 **Question**

Write a program which takes a value from user and then using for loop display all of its divisors.

SANAULLAH KHAN
MSc. MATHEMATICS
FOR QUIZ CONTECT
(03066793776)

YOUTUBE CHANNEL: S KHAN ACADEMY

CODE:

```
clc
clear all
close all
x=input('Enter a number:')
k=x;
for i=1:k;
if mod(x,i)==0
disp(i)
end
end
```

9 **Question**

Numerically approximate the solution of the following ode dy/dt=2t use time interval [0 5] and initial condition is y(0)=0 by using ode45 command.

CODE:

```
clc
clear all
close all
f=@(t,y)2*t;
interval=[0 5];
y0=0;
[t,y]=ode45(f,interval,y0)
plot(t,y)
```

10 **Question**

Write a function with name 'myfunction' which takes any two number as input and returns the greater number.

SANAULLAH KHAN
MSc. MATHEMATICS
FOR QUIZ CONTECT
(03066793776)

YOUTUBE CHANNEL: S KHAN ACADEMY

CODE:

```
function[x,y]=myfunction(a,b)
if a>b
disp('The max number is a')
else
disp('The max number is b')
end
```

11 **Question**

Write a function with name 'newfunction' which takes any two number as input and returns the highest common factor of the number.

CODE:

```
function[HCF]=newfunction(a,b)
for k=1:1:(a+b);
    if mod(a,k)==0 && mod(b,k)==0
        HCF=k;
end
end
```

12 **Question**

Write a program which prints all the numbers divisible by 3 and 5 for a given number.

CODE:

```
clc
clear all
close all
x=input('Enter The number:')
n=x;
for k=1:n;
    if mod(k,3)==0 && mod(k,5)==0
        disp(k)
end
end
```

SANAULLAH KHAN
MSc. MATHEMATICS
FOR QUIZ CONTECT
(03066793776)

YOUTUBE CHANNEL: S KHAN ACADEMY

13 **Question**

Write a for loop which adds first 30 odd numbers.

CODE:

```
clear all
close all
clc
i=1:60;
sum=0;
for k=1:2:length(i);
sum=sum+k;
end
disp(sum)
```

14 **Question**

Write a function name cube takes a number as an input if the number is even return

Cube of that number if odd then return square.

CODE:

```
function[cube square]=cube(n)
if rem(n,2)==1
square=n^2
else
cube=n^3
end
end
```

15 **Question**

Find sum and write program $\sum_{k=1}^{n}\left(\frac{3}{x}\right)^{k}$ and input x=4 and n=12.

SANAULLAH KHAN
MSc. MATHEMATICS
FOR QUIZ CONTECT
(03066793776)

YOUTUBE CHANNEL: S KHAN ACADEMY

CODE:

```
clc
clear all
close all
n=12;
x=4;
sum=0;
for k=1:n
sum=sum+(3/x)^k;
end
disp(sum)
```

16 **Question**

Solve by using the newton Raphson's method $x^3 + x$ by using for loop when x0=2 and n=3.

CODE:

```
clc
clear all
close all
f=@(x)x^3+x
df=@(x)3*x^2+1
x0=2;
n=3;
x=x0;
for i=1:n;
x=x-f(x)/df(x)
end
```

17 **Question**

Solve by using the newton Raphson's method $x^3 + x$ by using for loop when x0=2 and n=3.

SANAULLAH KHAN
MSc. MATHEMATICS
FOR QUIZ CONTECT
(03066793776)

CODE:

```
clc
clear all
close all
f=@(x)x^3+x
df=@(x)3*x^2+1
x0=2;
n=3;
x=x0;
tol=10^-4
while abs(f(x))>tol
x=x-f(x)/df(x)
end
fprintf('Root: %o.15 f\n',x)
```

18 **Question**

Write a function with the name 'practisefunction' which takes any three random variables as an input and returns the number in ascending order.

CODE:

```
function [Ascending]=practisefunction(a,b,c)
A=[a,b,c]
Ascendingorder=sort(A)
end
```

19 **Question**

Find factorial of a number by using while loop.

CODE:

```
clc
clear all
close all
```

SANAULLAH KHAN
MSc. MATHEMATICS
FOR QUIZ CONTECT
(03066793776)

```
n=input('enter a number:');
i=n;
fact=1;
if i==0
fact=1;
else if i<0
disp('the number you are entered is nagative')
   end
end
while i>0
fact=fact*i;
i=i-1;
end
disp(fact)
```

20 **Question**

Find factorial of a number by using for loop.

CODE:

```
clc
clear all
close all
n=input('Enter a number:');
fact=1;
for i=1:n;
fact=fact*i;
end
disp(fact)
```

21 **Question**

write a function with name examquestions which take any two number as an input and return the sum and product of those number as output.

SANAULLAH KHAN
MSc. MATHEMATICS
FOR QUIZ CONTECT
(03066793776)

YOUTUBE CHANNEL: S KHAN ACADEMY

CODE:

```
function[a b]=examquestions(a,b)
sum=a+b
product=a*b
end
```

22 **Question**

Use while loop to calculate the factorial of 10.

CODE:

```
clc
clear all
close all
n=10;
i=n;
fact=1;
while i>0
fact=fact*i;
i=i-1;
end
disp(fact)
```

23 **Question**

Numerically solve the following equation by using ode45 command

$\frac{dy}{dx} = \frac{y}{t^2} + \frac{2}{t}$ $y(0) = 1$ $[2, -2]$

CODE:

```
clc
clear all
close all
f=@(t,y)(y/t^2+2/t);
interval=[-2 2];
y0=1;
[t,y]=ode45(f,interval,y0)
```

SANAULLAH KHAN
MSc. MATHEMATICS
FOR QUIZ CONTECT
(03066793776)

```
plot(t,y)
```

24 **Question**

$$A = \begin{bmatrix} \frac{3}{11} & \frac{2}{5} & \frac{4}{7} \\ \frac{2}{3} & \frac{4}{5} & \frac{6}{5} \end{bmatrix} \quad \& \quad B = \begin{bmatrix} \frac{1}{3} & \frac{8}{4} & 4 \\ \frac{2}{9} & \frac{7}{3} & \frac{9}{6} \\ 6 & \frac{7}{4} & \frac{5}{7} \end{bmatrix}$$

Find

1. **Inv(A)**
2. **A*B**

CODE:

```
clc
clear all
close all
A=sym([3/11 2/5 4/7;2/3 4/5 6/5]);
B=sym([1/3 8/4 4;2/9 3/7 9/6;6 7/4 5/7]);
C=inv(B)
D=sym(A*B)
```

25 **Question**

$2x + y + z = 2$, $-x + y - z = 3$, $x + 2y + 3z = -10$

1. **Write system in MATLAB as an augmented matrix**
2. **Make reduced echelon form**

CODE:

```
clc
clear all
close all
A=[2,1,1;-1,1,-1;1,2,3]
b=[2;3;-10]
```

SANAULLAH KHAN
MSc. MATHEMATICS
FOR QUIZ CONTECT
(03066793776)

YOUTUBE CHANNEL: S KHAN ACADEMY

```
Ab=[A b]
%AX=B%
C=rref(Ab)
```

26 **Question**

$y(x) = \sqrt{1+x^3}$ **solve the equation using Simpson's 3/8 rule interval [1 5] and keep 12 sub interval.**

CODE:

```
clc
clear all
close all
f=@(x)sqrt(1+x^3);
a=1;
b=5;
n=12;
% if step size is given the we usw n=(b-a)/h
% if subinterval is given the we use h=(b-a)/n
h=(b-a)/n;
sum1=0;
sum2=0;
for i=1:n-1;
 x(i)=a+i*h;
 if rem(i,3)==0
 sum1=sum1+f(x(i));
 else
 sum2=sum2+f(x(i));
 end
end
Area=3*h/8*(f(a)+f(b)+2*sum1+3*sum2)
```

27 **Question**

$\frac{d^2y}{dx^2} + 2\frac{dy}{dx} + 5y = 7$ and $y(0) = 1$ and $y'(0) = 0$

Solve the differential equation.

SANAULLAH KHAN
MSc. MATHEMATICS
FOR QUIZ CONTECT
(03066793776)

YOUTUBE CHANNEL: S KHAN ACADEMY

CODE:

```
clc
clear all
close all
syms y(t)
Dy=diff(y,t)
ode=diff(y,t,2)+2*diff(y,t)+5*y==7
cond1=y(0)==1
cond2=Dy(0)==0
cond=[cond1,cond2]
sol=dsolve(ode,cond)
```

28

Question

Solve $x - 2\sin(x^2) = 0$ **by using bisection method perform 10 iterations to find the root between 0.4 and 0.6.**

CODE:

```
clc
clear all
close all
f=@(x)x-2*sin(x^2)
a=0.4;
b=0.6;
n=10;
if f(a)*f(b)>0
 disp('there is no change of sign:')
 return
end
tol=10^-4;
disp('iteration number a b c')
i=1;
while (abs(a-b)>=tol)
 c=(a+b)/2;
```

SANAULLAH KHAN
MSc. MATHEMATICS
FOR QUIZ CONTECT
(03066793776)

```
if f(c)==0;
fprintf('root of the function is %f:',c)
return
end
fprintf('%2i \t %f \t %f \t %f \n',i,a,b,c)
if f(a)*f(c)>0
a=c;
else
b=c;
i=i+1;
end
end
m=(a+b)/2
fprintf('root lies at c=%f \n',m)
```

29 **Question**

Write a function with name 'circlecalculation' which take the circle of radius as input and circf as output.

CODE:

```
function [circf]=circlecalculation(radius)
radius=input('Enter the radius of circle:')
circf=2*pi*radius
end
```

30 **Question**

$y(x) = \sqrt{1 + x^3}$ solve the equation using Simpson's 1/8 rule interval [1 5] and keep 8 sub interval.

CODE:

```
clc
clear all
close all
f=@(x)sqrt(1+x^3)
a=1;
```

SANAULLAH KHAN
MSc. MATHEMATICS
FOR QUIZ CONTECT
(03066793776)

```
b=5;
n=8;
% if step size is given the we usw n=(b-a)/h
% if subinterval is given the we use h=(b-a)/n
h=(b-a)/n;
sumofeven=0;
sumofodd=0;
for k=1:n-1;
 x(k)=a+k*h;
 if rem(k,2)==0
 sumofeven=sumofeven+f(x(k));
 else
 sumofodd=sumofodd+f(x(k));
 end
end
Area=h/3*(f(a)+f(b)+2*sumofeven+4*sumofodd)
```

31 **Question**

$$A = \begin{bmatrix} \frac{3}{11} & \frac{2}{5} & \frac{4}{7} \\ \frac{2}{3} & \frac{4}{5} & \frac{6}{5} \end{bmatrix} \quad \& \quad B = \begin{bmatrix} \frac{1}{3} & \frac{8}{4} & 4 \\ \frac{2}{9} & \frac{7}{3} & \frac{9}{6} \\ 6 & \frac{7}{4} & \frac{5}{7} \end{bmatrix}$$

Find

1. **Sum of 3rd row of B**
2. **Sum of 2nd column of A**

CODE:

```
clc
clear all
close all
A=sym([3/11 2/5 4/7;2/3 4/5 6/5])
B=sym([1/3 8/4 4;2/9 3/7 9/6;6 7/4 5/7])
SumA=sum(A(:,2))
```

SANAULLAH KHAN
MSc. MATHEMATICS
FOR QUIZ CONTECT
(03066793776)

```
SumB=sum(B(3,:))
```

32 **Question**

Solve the following equation $\frac{X^2}{2}$ by using trapezoidal rule in the interval [4 6] by using 10 sub interval.

CODE:

```
clc
clear all
close all
f=@(x)x^2/2
a=4;
b=6;
n=10;
% if step size is given the we usw n=(b-a)/h
% if subinterval is given the we use h=(b-a)/n
h=(b-a)/n;
sum=0;
for i=1:n-1;
 x(i)=a+i*h;
 y(i)=f(x(i));
 sum=sum+y(i);
end
Area=h/2*(f(a)+f(b)+2*sum)
```

33 **Question**

Solve the following equation $\frac{X^2}{2}$ by using Reimann sum in the interval [4 6] by using 10 sub interval.

CODE:

```
clc
clear all
```

```
close all
f=@(x)x^2/2
a=4;
b=6;
n=10;
h=(b-a)/n;
value=0;
for i=1:n;
 c=a+i*h;
 value=value+f(c);
end
Area=value*h
```

34 **Question**

Find the value of X if

$A = \begin{bmatrix} 1 & 3 \\ 2 & 4 \end{bmatrix} \quad and \quad B = \begin{bmatrix} -2 \\ 3 \end{bmatrix}$

CODE:

```
clc
clear all
close all
A=[1,3;2,4]
B=[-2;3]
% As we know that
% AX=B
% X=inv(A)*B
X=inv(A)*B
```

35 **Question**

Solve the system of linear equation by using solve command.

$3x + 2y + x = 1 \quad , \quad y = 2 \quad , \quad x + 2y = 3$

CODE:

```
clc
```

SANAULLAH KHAN
MSc. MATHEMATICS
FOR QUIZ CONTECT
(03066793776)

```
clear all
close all
eqn1='3*x+2*y+z=1';
eqn2='y=2';
eqn3='x+2*y=3';
sol=solve(eqn1,eqn2,eqn3)
Answer=double([sol.x;sol.y;sol.z])
```

36

Question

Find the eigen values and eigen vector of the following matrix.

$$A = \begin{bmatrix} 1 & 3 & 4 \\ 7 & 8 & 9 \\ 3 & 1 & 3 \end{bmatrix}$$

CODE:

```
clc
clear all
close all
A=[3,1,4;7,8,9;3,1,3]
V=eig(A)
[P D]=eig(A)
```

SANAULLAH KHAN
MSc. MATHEMATICS
FOR QUIZ CONTECT
(03066793776)